

MUHAMMAD

SOFTWARE ENGINEER

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• LinkedIn Profile Link: [MUHAMMAD](#) • Portfolio: [Link](#)

SUMMARY

Software Engineer specializing in Built and maintained of backend APIs and AI agents. Experienced in Fullstack development and proficient in DevOps practices, including Docker, Kubernetes, Cloud computing and Ubuntu/Linux server administration. Additionally skilled in data wrangling and exploratory data analysis (EDA). Recognized with the Rising Star Award under the Governor Sindh IT Initiative for outstanding contributions to AI-driven automation and innovation.

WORK EXPERIENCE

COMBINE FOUNDATION

May-July 2025

The NPO (FullStack Developer)

- Developed and maintained scalable backend systems using FastAPI, implementing secure JWT authentication and well-structured RESTful APIs.
- Designed and built a full-stack web application using Next.js with SSR, dynamic routing, and real-time data handling, integrating frontend and backend seamlessly.
- Engineered and managed relational databases with Alembic for schema migrations, ensuring efficient data handling and optimized performance.
- Deployed applications with backend on Railway and frontend on Vercel, ensuring reliable and production-ready system availability.
- Developed complete e-commerce platform using Next.js integrated with Sanity CMS, including secure user authentication and management via Clerk.

ITVE Pvt. Ltd.

2026

Education and Tech Co. (Backend Developer)

- Developed and managed RESTful APIs using FastAPI for a multi-role application supporting donors, students, and general users.
 - Implemented secure JWT-based authentication and authorization to manage rolebased access control.
 - Designed and maintained MongoDB databases, ensuring efficient data modeling, storage, and retrieval.
 - Built scalable backend architecture for multi-user systems, ensuring smooth data flow and containerized the application using Docker for consistent and reliable deployment.
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EDUCATION

Bachelors in Software Engineering

2023-2027 (Present)

- Sindh Madressatul Islam University
- GPA: 3.2

ADDITIONAL PROJECTS & SKILLS

- Knowledge of Retrieval-Augmented Generation (RAG) architectures using LangGraph, with vector embeddings, semantic search, document retrieval, agentic workflows, and LLM-based context-aware response generation.
 - Managed and configured Ubuntu/Linux servers, including system updates, service management, user access control, Bash Scripting and network configuration.
 - Deployed and managed containerized applications using Docker and orchestrated them with Kubernetes, implementing CI/CD pipelines with Jenkins for automation.
 - Performed manual and automated testing using Selenium to ensure application reliability, functionality, and cross-browser compatibility.
 - Conducted Exploratory Data Analysis (EDA) and data preprocessing, including handling missing values, feature scaling, and outlier detection.
 - Applied data analytics skills from Google play store, including data cleaning, visualization, and analytical problem-solving with a focus on data ethics.
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TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, SQL
 - **Frameworks & Technologies:** FastAPI, Next.js, Tailwind CSS, Open AI Agents SDK, MCP (Servers), Selenium, n8n
 - **Databases:** MySQL, PostgreSQL, MongoDB
 - **Cloud & DevOps Tools:** Docker, Kubernetes, Jenkins, Bash Scripting
 - **Data Analysis & Visualization:** Pandas, NumPy, Power BI, Jupyter Notebook, Google Colab, Matplotlib, Seaborn
 - **Operating Systems & Server Administration:** Ubuntu/Linux, SSH Configuration, Firewall Setup
 - **Software Engineering:** Software Design & Architecture, Requirement Gathering, Project Management, Agile Methodology
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CERTIFICATE

2023-2025

Governor's Initiative for Artificial Intelligence & Computing (GIAIC)

- Governor Sindh Initiative for GenAI, Web3, and Metaverse
- Link: [GIAIC](#)